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1  #!/usr/bin/env python3
2  # -*- coding: utf-8 -*-
3  """
4  Example 7.2 Polynomial Features
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6  """
7
8  import IPython as IP
9  IP.get_ipython().run_line_magic('reset', '-sf')
10
11 import numpy as np
12 import matplotlib.pyplot as plt
13 import sklearn as sk
14 from sklearn import datasets
15 from sklearn import pipeline
16 from sklearn import svm
17
18 plt.close('all')
19
20 %% Build and plot the data
21
22 # build the data
23 X, y = sk.datasets.make_moons(n_samples=100, noise=0.25, random_state=2)
24
25 plt.figure()
26 plt.plot(X[:,0][y==0], X[:,1][y==0], 's')
27 plt.plot(X[:,0][y==1], X[:,1][y==1], 'd')
28 plt.xlabel("$x_1$")
29 plt.ylabel("$x_2$")
30
31 %% SVM polynomial features
32 svm_clf = sk.pipeline.Pipeline([
33     ("poly_features", sk.preprocessing.PolynomialFeatures(degree=3)),
34     ("scaler", sk.preprocessing.StandardScaler()),
35     ("svm_clf", sk.svm.LinearSVC(C=10))
36 ])
37 svm_clf.fit(X, y)
38
39 # make the 2d space for the color
40 x1 = np.linspace(-2, 3, 200)
41 x2 = np.linspace(-2, 2, 100)
42 x1_grid, x2_grid = np.meshgrid(x1, x2)
43
44 # calculate the binary decions and predecision values
45 X2 = np.vstack((x1_grid.ravel(), x2_grid.ravel())).T
46 y_pred = svm_clf.predict(X2).reshape(x1_grid.shape)
47 y_decision = svm_clf.decision_function(X2).reshape(x1_grid.shape)
48
49 con_lines = [-30,-20,-10,-5,-2,-1,0,1,2,5,10,20,30]
50
51
52 # plot the figure
53 plt.figure()
54 # provide the solid background color for classification
55 plt.contourf(x1_grid, x2_grid, y_pred, cmap=plt.cm.brg, alpha=0.2)
56 # add the contour colors for the threshold
57 plt.contourf(x1_grid, x2_grid, y_decision, con_lines, cmap=plt.cm.brg, alpha=0.1)
58 # add the contour lines
59 contour = plt.contour(x1_grid, x2_grid, y_decision, con_lines, cmap=plt.cm.brg)
60 plt.clabel(contour, inline=1, fontsize=12)
61 plt.plot(X[:, 0][y==0], X[:, 1][y==0], "s")
62 plt.plot(X[:, 0][y==1], X[:, 1][y==1], "d")
63 plt.xlabel("$x_1$")
64 plt.ylabel("$x_2$")

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